Lecture 8. Numerical Integration

Newton–Cotes rules, Gaussian quadrature, orthogonal polynomials, Laplace approximation

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We want $\int_a^b f(x)\,dx$. Integrals are everywhere in statistics but rarely have closed forms:

- expectations, $E\sin X = \int \sin(x) f_X(x)\,dx$;
- Bayesian normalizing constants $\int f(x \mid \theta)\pi(\theta)\,d\theta$;
- likelihoods of mixed models, e.g. a Poisson GLM with random intercept, $\prod_i \int \prod_j f(y_{ij} \mid \gamma_i)\phi(\gamma_i)\,d\gamma_i$.

Two approaches: quadrature (now) and Monte Carlo (Lecture 10).

8.1 Newton–Cotes rules

Split $[a, b]$ into n intervals of length $h = (b-a)/n$, $x_i = a + ih$, approximate f on each $[x_i, x_{i+1}]$ by an interpolating polynomial of degree m, integrate the polynomial exactly, and add up.

- $m = 0$, **Riemann**: $\int_{x_i}^{x_{i+1}} f \approx h f(x_i)$.
- $m = 1$, **trapezoidal**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{2}\{f(x_i) + f(x_{i+1})\}$, so $T(n) = h\{\frac{1}{2}f(a) + f(x_1) + \cdots + f(x_{n-1}) + \frac{1}{2}f(b)\}$.
- $m = 2$, **Simpson**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{6}\{f(x_i) + 4f(\frac{x_i + x_{i+1}}{2}) + f(x_{i+1})\}$.

Proposition 8.1 (Accuracy)

For smooth f, the total errors are $O(n^{-1})$ (Riemann), $O(n^{-2})$ (trapezoidal) and $O(n^{-4})$ (Simpson). More precisely, per subinterval the local errors are

$$\frac{h^2}{2}f'(x_i) + O(h^3), \qquad -\frac{h^3}{12}f''(x_i) + O(h^4), \qquad -\frac{h^5}{2880}f^{(4)}(x_i) + O(h^6).$$

Proof. Integrate the Taylor expansion $f(x) = f(x_i) + (x - x_i)f'(x_i) + \frac{1}{2}(x - x_i)^2 f''(x_i) + \dots$ over $[x_i, x_i + h]$:

$$\int_{x_i}^{x_{i+1}} f = h f(x_i) + \frac{h^2}{2}f'(x_i) + \frac{h^3}{6}f''(x_i) + \frac{h^4}{24}f^{(3)}(x_i) + \frac{h^5}{120}f^{(4)}(x_i) + O(h^6).$$

Riemann: the error is $\frac{h^2}{2}f'(x_i) + O(h^3)$. *Trapezoid*: expanding $f(x_{i+1})$ gives $\frac{h}{2}\{f(x_i) + f(x_{i+1})\} = hf(x_i) + \frac{h^2}{2}f' + \frac{h^3}{4}f'' + O(h^4)$, so the error is $(\frac{1}{6} - \frac{1}{4})h^3 f'' + O(h^4) = -\frac{h^3}{12}f''(x_i) + O(h^4)$. *Simpson*: expanding f at the midpoint and endpoint, the rule matches the terms through h^4 and differs at h^5 by $(\frac{1}{120} - \frac{5}{576})h^5 f^{(4)} = -\frac{h^5}{2880}f^{(4)}$. Summing n local errors multiplies by $n = (b-a)/h$, giving global orders h, h^2, h^4. ∎

Simpson uses about twice as many function evaluations per interval, so a fair comparison fixes the total number of evaluations; Simpson still wins by orders of magnitude for smooth f. For $\int_0^2 e^{-x}dx = 1 - e^{-2} \approx 0.864665$, with $n = 10$ the three rules have errors 8.9×10^{-2}, 2.9×10^{-3} and 4.8×10^{-7}, and on a log–log plot of error against n the slopes are -1, -2 and -4.

Example 8.1 (An infinite range)

Newton–Cotes rules need a finite interval. For $\int_1^\infty e^{-x}x^{-1/2}dx$ substitute $t = 1/x$, $dx = -dt/t^2$:

$$\int_1^\infty e^{-x}x^{-1/2}dx = \int_0^1 e^{-1/t}t^{-3/2}\,dt.$$

The new integrand cannot be evaluated at $t = 0$ (it is $0 \cdot \infty$ in floating point, giving `NaN`), so the trapezoidal and Simpson rules break at the endpoint. But $e^{-1/t} t^{-3/2} \to 0$ as $t \downarrow 0$, together with all its derivatives, so defining it as 0 at $t = 0$ gives a smooth integrand on $[0, 1]$ and the rules converge at their usual rates, to $\sqrt{\pi}\,\mathrm{erfc}(1) = \Gamma(\frac{1}{2}, 1) \approx 0.278806$. (Alternatively, use a rule that does not evaluate the endpoints, such as the midpoint rule or Gaussian quadrature.)

General m: interpolation. With $m+1$ distinct nodes $x_0, \ldots, x_m$, the **Lagrange polynomials** $L_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}$ satisfy $L_j(x_k) = \delta_{jk}$, so $p(x) = \sum_j y_j L_j(x)$ interpolates (x_j, y_j).

Theorem 8.2 (Polynomial interpolation)

For $m + 1$ distinct points there is a unique polynomial of degree at most m through $(x_0, y_0), \ldots, (x_m, y_m)$.

Proof. Existence is the Lagrange form. If q is another, $p - q$ has degree $\leq m$ and $m + 1$ roots, hence is identically 0. ■

Corollary 8.3

If f is a polynomial of degree at most m, then $\int_a^b f = \sum_j w_j f(x_j)$ with $w_j = \int_a^b L_j(x)\,dx$.

Proof. By uniqueness, $f = \sum_j f(x_j) L_j$; integrate. ■

8.2 Gaussian quadrature

Newton–Cotes fixes equally spaced nodes and chooses weights to integrate polynomials of degree m exactly. Choosing the nodes as well gains much more.

Polynomials of degree $\leq m$ on (a, b) form a vector space $\mathcal{P}^m$ with inner product

$$\langle p, q \rangle = \int_a^b p(x) q(x) w(x)\,dx$$

for a positive weight w (here $w \equiv 1$). Gram–Schmidt applied to $1, x, x^2, \ldots$ produces orthonormal polynomials $q_0, q_1, \ldots$, with q_k of exact degree k and orthogonal to all of $\mathcal{P}^{k-1}$; for $w \equiv 1$ on $(-1, 1)$ these are the **Legendre polynomials**.

Lemma 8.4

The roots of q_k are real, simple and lie in (a, b).

Proof. Let $r_1 < \cdots < r_l$ be the points in (a, b) where q_k changes sign. If $l < k$, then $q_k(x) \prod_{i \leq l} (x - r_i)$ does not change sign on (a, b) and is not identically zero, so its integral is nonzero. But $\prod_{i \leq l} (x - r_i)$ has degree $l < k$ and is orthogonal to q_k: a contradiction. So q_k has k sign changes, hence k distinct real roots in (a, b). ■

Theorem 8.5 (Gaussian quadrature)

Let $x_0, \dots, x_m$ be the roots of q_{m+1} and $w_j = \int_a^b L_j(x)w(x)\,dx$. Then

$$\int_a^b f(x)w(x)\,dx = \sum_{j=0}^{m} w_j f(x_j)$$

for every polynomial f of degree at most $2m+1$, and all $w_j > 0$. Degree $2m+1$ cannot be improved.

Proof. Divide: $f = gq_{m+1} + r$ with $g, r \in \mathcal{P}^m$. Then $\int gq_{m+1}w = \langle g, q_{m+1}\rangle = 0$, so $\int fw = \int rw$. At the nodes $q_{m+1}(x_j) = 0$, so $f(x_j) = r(x_j)$, and by Corollary 8.3 (with weight w) $\int rw = \sum w_j r(x_j) = \sum w_j f(x_j)$. Positivity: L_k^2 has degree $2m$, so $0 < \int L_k^2 w = \sum_j w_j L_k(x_j)^2 = w_k$. Optimality: $f = \prod_j (x - x_j)^2$ has degree $2m+2$ and $\int fw > 0$, while the rule gives 0. ■

So $m+1$ well-chosen nodes integrate polynomials of degree $2m+1$ exactly, against degree m for equally spaced nodes.

Weight functions. Changing the weight gives different families of orthogonal polynomials, suited to different integrals $\int g(x)w(x)\,dx$:

polynomials	weight $w(x)$	domain	related distribution
Legendre	1	$(-1, 1)$	uniform
Laguerre	$x^\alpha e^{-x}$	$(0, \infty)$	gamma
Hermite	e^{-x^2}	$(-\infty, \infty)$	normal
Jacobi	$(1-x)^\alpha(1+x)^\beta$	$(-1, 1)$	beta (after rescaling)

Each family satisfies a three-term recursion $q_m(x) = (\alpha_m + \beta_m x)q_{m-1}(x) - \gamma_m q_{m-2}(x)$, and its roots *interlace* (the roots of q_{m+1} separate those of q_m). The nodes are computed by Newton's method using these properties (or as eigenvalues of a tridiagonal matrix).

Gauss–Hermite quadrature. $\int f(x)\,dx = \int \{f(x)e^{x^2}\}e^{-x^2}dx \approx \sum_j w_j f(x_j)e^{x_j^2}$. It works well only if $f(x)e^{x^2}$ is close to a polynomial. For $f(x) = e^{-(x-\mu)^2/(2\sigma^2)}$ with μ far from 0 or σ^2 far from $\frac{1}{2}$, the nodes sit where f has no mass and the rule fails. The remedy is to *centre and scale*: substituting $z = (x-\mu)/(\sqrt{2}\sigma)$ gives $\sqrt{2}\sigma \int e^{-z^2}dz$, which a single node integrates exactly.

Example 8.2 (Binomial–logit posterior)

$x \sim B(n, p)$ with $p = e^\theta/(1+e^\theta)$ and prior $\theta \sim N(0, \sigma^2)$. The posterior normalizing constant is $\int \bar{g}(\theta)\,d\theta$ with $\bar{g}(\theta) = e^{\theta x}(1+e^\theta)^{-n}e^{-\theta^2/(2\sigma^2)}$. Applying Gauss–Hermite after only rescaling by σ performs badly when $x/n = 0.9$, because the integrand peaks near $\theta = \log 9 = 2.2$, far from the nodes. Centring at the maximizer θ^* of $\bar{g}$ and writing $\int \bar{g} = \int \{\bar{g}(z+\theta^*)e^{z^2}\}e^{-z^2}dz$ gives an accurate rule with few nodes.

Example 8.3 (Poisson posterior: centre *and* scale)

Let $y_1, \dots, y_{10}$ be Poisson with mean e^α, prior $\alpha \sim N(0, 100)$, and data $11, 19, 27, 12, 14, 11, 10, 13, 15, 10$ ($\sum y_i = 142$). The log unnormalized posterior is $h(\alpha) = 142\alpha - 10e^\alpha - \alpha^2/200$. Its mode $\hat{\alpha} = 2.65306$ solves $h' = 0$ (by Newton or bisection), and the

curvature gives the scale $\hat{s} = \{-h''(\hat{\alpha})\}^{-1/2} = 0.08392$. Substituting $\alpha = \hat{\alpha} + \sqrt{2}\hat{s}z$,

$$E[g(\alpha) \mid y] = \frac{\int g(\hat{\alpha} + \sqrt{2}\hat{s}z) e^{h(\hat{\alpha}+\sqrt{2}\hat{s}z) - h(\hat{\alpha}) + z^2} e^{-z^2} dz}{\int e^{h(\hat{\alpha}+\sqrt{2}\hat{s}z) - h(\hat{\alpha}) + z^2} e^{-z^2} dz},$$

and both integrals go to Gauss–Hermite. (The constants $\sqrt{2}\hat{s}$ cancel; subtracting $h(\hat{\alpha})$ prevents overflow of e^h, since $h(\hat{\alpha}) \approx 235$.) Since the posterior is nearly normal, the bracketed functions are nearly constant or polynomial:

nodes	posterior mean	posterior variance
2	2.651881	0.0070417
5	2.649534	0.0070677
10	2.6495297	0.00706793
15	2.649529692	0.007067929

Fifteen nodes reproduce adaptive numerical integration to 10 significant digits. Without centring, the nodes of e^{-z^2} sit around 0, where the posterior (concentrated near 2.65 with standard deviation 0.084) is numerically zero, and the rule fails completely.

Example 8.4 (Variance of the sample median)

For $X_1, \ldots, X_n$ i.i.d. $N(0,1)$, $n = 2m+1$, the median $X_{(m+1)}$ has density $n\binom{n-1}{m}\Phi(x)^m\{1-\Phi(x)\}^m\phi(x)$: one observation falls in $[x, x+dx)$ (n choices, probability $\phi(x)dx$), and exactly m of the other $n-1$ fall below x. The density is symmetric, so $EX_{(m+1)} = 0$ and

$$\begin{aligned}\operatorname{Var} X_{(m+1)} &= \int_{-\infty}^{\infty} x^2 n \binom{n-1}{m} \Phi(x)^m \{1-\Phi(x)\}^m \phi(x)\, dx \\ &= \frac{2n}{\sqrt{\pi}} \binom{n-1}{m} \int u^2 \Phi(\sqrt{2}u)^m \{1-\Phi(\sqrt{2}u)\}^m e^{-u^2} du,\end{aligned}$$

after $x = \sqrt{2}u$. Gauss–Hermite with k nodes gives:

k	2	4	8	16	32	64	exact	$\frac{\pi}{2n}$
$n = 11$	0.117	0.238	0.234	0.161	0.1379	0.13716	0.13716	0.1428
$n = 21$	0.007	0.057	0.127	0.123	0.0840	0.07349	0.07326	0.0748

Convergence is slow and erratic because $\Phi^m(1-\Phi)^m$ is sharply peaked, with width of order $n^{-1/2}$, while the Hermite nodes are spread for a unit-variance shape. Rescaling by the asymptotic standard deviation, as in the previous example, fixes this. The asymptotic variance $1/\{4n\phi(0)^2\} = \pi/(2n)$ overstates the exact value by 4.1% for $n = 11$ and 2.1% for $n = 21$.

Example 8.5 (Generalized Gauss–Laguerre)

With $x = u^2$, $\int_0^\infty \sqrt{x}e^{-x}dx = 2\int_0^\infty u^2 e^{-u^2}du = \frac{\sqrt{\pi}}{2}$ $(= \Gamma(\frac{3}{2}))$. To compute $\int_0^\infty f(x)\sqrt{x}e^{-x}dx$, the natural rule is Gauss quadrature with weight $w(x) = x^{1/2}e^{-x}$ (generalized Laguerre, $\alpha = \frac{1}{2}$), whose nodes are the roots of the corresponding orthogonal polynomial. For $f \equiv 1$ a single node ($m = 0$, weight $w_0 = \Gamma(\frac{3}{2})$) is already exact. For $f = \log x$, with exact value $\Gamma'(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}(2 - \gamma - 2\log 2) \approx 0.032338$, two sources of error remain.

(i) *Method error*: $\log x$ is not a polynomial and is singular at 0, so the rule is not exact and the error decreases only slowly (algebraically, not geometrically) with m.

(ii) *Floating-point error*: the nodes and weights are themselves computed approximately (by Newton's method or an eigenvalue problem), the smallest node is close to 0 where $\log x$ is large, the largest weights are tiny and may underflow for large m, and the result is small (0.03) compared with the terms being summed, so cancellation between positive and negative terms magnifies rounding errors.

8.3 Laplace approximation

For integrals $\int f(x)e^{-ag(x)}dx$ with large a, where g has a unique minimum at 0, the integrand concentrates near 0, and $g(x) \approx g(0) + \frac{1}{2}g''(0)x^2$ gives

$$\int f(x)e^{-ag(x)}dx \approx f(0)e^{-ag(0)}\sqrt{\frac{2\pi}{ag''(0)}}:$$

integration is replaced by optimization.

Theorem 8.6 (Laplace)

Suppose (1) for every $\delta > 0$ there is $\rho > 0$ with $g(x) - g(0) \geq \rho$ for $|x| \geq \delta$;
(2) g is twice continuously differentiable near 0 with $g'(0) = 0$, $g''(0) > 0$;
(3) f is continuous near 0 with $f(0) > 0$;
(4) $\int |f|e^{-ag} < \infty$ for $a \geq a_0$. Then

$$\lim_{a\to\infty} \frac{\int f(x)e^{-ag(x)}dx}{f(0)e^{-ag(0)}\sqrt{2\pi/(ag''(0))}} = 1.$$

Proof. Take $g(0) = 0$. Given $\epsilon \in (0, \frac{1}{2}g''(0))$, choose δ such that for $|x| < \delta$,

$$(1-\epsilon)f(0) \leq f(x) \leq (1+\epsilon)f(0), \qquad \tfrac{1}{2}(g''(0)-2\epsilon)x^2 \leq g(x) \leq \tfrac{1}{2}(g''(0)+2\epsilon)x^2.$$

Tails. For $|x| \geq \delta$, $g \geq \rho$, so for $a \geq a_0$,

$$\int_{|x|\geq\delta} |f|e^{-ag} = \int_{|x|\geq\delta} |f|e^{-a_0 g}e^{-(a-a_0)g} \leq e^{-(a-a_0)\rho}\int |f|e^{-a_0 g} = O(e^{-\rho a}).$$

Centre, upper bound. $\int_{|x|<\delta} fe^{-ag} \leq (1+\epsilon)f(0)\int e^{-\frac{a}{2}(g''(0)-2\epsilon)x^2}dx = (1+\epsilon)f(0)\sqrt{\frac{2\pi}{a(g''(0)-2\epsilon)}}$. *Centre, lower bound.* $\int_{|x|<\delta} fe^{-ag} \geq (1-\epsilon)f(0)\int_{|x|<\delta} e^{-\frac{a}{2}(g''(0)+2\epsilon)x^2}dx$, and the Gaussian integral over $|x| \geq \delta$ is $O(e^{-\lambda a})$ with $\lambda = \frac{1}{2}(g''(0)+2\epsilon)\delta^2$. Multiplying by $\sqrt{a}$ and letting $a \to \infty$,

$$(1-\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)+2\epsilon}} \leq \liminf \sqrt{a}\int fe^{-ag} \leq \limsup \sqrt{a}\int fe^{-ag} \leq (1+\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)-2\epsilon}}.$$

Let $\epsilon \to 0$. ■

Example 8.6 (Stirling's formula)

With $x = tz$, $\Gamma(t+1) = \int_0^\infty x^t e^{-x}dx = t^{t+1}\int_0^\infty e^{-t(z-\log z)}dz$. Here $g(z) = z - \log z$ is minimized at $z = 1$ with $g(1) = 1$, $g''(1) = 1$, so $\Gamma(t+1) \sim \sqrt{2\pi}\, t^{t+1/2}e^{-t}$.

Posterior expectations. With log-likelihood $\ell_n(\theta)$, log-prior $\pi(\theta)$ and $\tilde{h} = \log h > 0$,

$$E[h(\Theta) \mid x] = \frac{\int e^{\tilde{h}+\ell_n+\pi} d\theta}{\int e^{\ell_n+\pi} d\theta}.$$

Apply Laplace to numerator and denominator separately, at the maximizers $\tilde{\theta}$ of $\tilde{h} + \ell_n + \pi$ and $\hat{\theta}$ of $\ell_n + \pi$ (the posterior mode):

$$E[h(\Theta) \mid x] \approx \exp\{(\tilde{h} + \ell_n + \pi)(\tilde{\theta}) - (\ell_n + \pi)(\hat{\theta})\} \sqrt{\frac{(\ell_n + \pi)''(\hat{\theta})}{(\tilde{h} + \ell_n + \pi)''(\tilde{\theta})}}.$$

For large n the log-posterior is sharply peaked, and the approximation is accurate (its relative error is $O(n^{-2})$, since the leading errors of numerator and denominator cancel).

Example 8.7 (Beta moments)

For $\Theta \sim \text{Beta}(\alpha, \beta)$ (a binomial posterior), $E\Theta^k = \frac{\alpha(\alpha+1)\cdots(\alpha+k-1)}{(\alpha+\beta)\cdots(\alpha+\beta+k-1)}$ exactly. With $g(\theta) = -(\alpha-1)\log\theta - (\beta-1)\log(1-\theta)$, minimized at $\hat{\theta} = \frac{\alpha-1}{\alpha+\beta-2}$,

$$\int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1} d\theta \approx \Big(\frac{\alpha-1}{\alpha+\beta-2}\Big)^{\alpha-1} \Big(\frac{\beta-1}{\alpha+\beta-2}\Big)^{\beta-1} \sqrt{\frac{2\pi(\alpha-1)(\beta-1)}{(\alpha+\beta-2)^3}},$$

and the same with α replaced by $\alpha + k$ for the numerator. If $\alpha, \beta \to \infty$ with $\alpha/(\alpha+\beta) \to p$, both the exact moment and its approximation tend to p^k.

Unlike quadrature, whose number of nodes grows exponentially with dimension, the Laplace approximation extends directly to $\mathbb{R}^d$: $\int f e^{-ag} \approx f(\hat{x}) e^{-ag(\hat{x})} (2\pi/a)^{d/2} \det\{\nabla^2 g(\hat{x})\}^{-1/2}$.